RPractical_Perez

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Part A: Basic Operations

A1

```
v <-seq(5, 50, by =5)
length(v)
```

```
## [1] 10
```

```
sum(v)
```

```
## [1] 275
```

```
mean(v)
```

```
## [1] 27.5
```

A2

```
w <-rep(c(1, 2, 3), times =4)
w
```

```
##  [1] 1 2 3 1 2 3 1 2 3 1 2 3
```

```
z <-rep(c(1, 2, 3), each =4)
z
```

```
##  [1] 1 1 1 1 2 2 2 2 3 3 3 3
```

A3

```r
17 %%5
```

```
## [1] 2
```

```r
17 /5
```

```
## [1] 3.4
```

```r
2 ^10
```

```
## [1] 1024
```

```r
sqrt(144)
```

```
## [1] 12
```

```r
round(pi, digits =3)
```

```
## [1] 3.142
```

A4

```r
a <-c(2, 4, 6, 8)
b <-c(1, 3, 5, 7)
a +b
```

```
## [1]  3  7 11 15
```

```r
a *b
```

```
## [1]  2 12 30 56
```

```r
a >b
```

```
## [1] TRUE TRUE TRUE TRUE
```

```r
sum(a *b)
```

```
## [1] 100
```

A5

```r
paste("Student-1", "Student-2", "Student-3")
```

```
## [1] "Student-1 Student-2 Student-3"
```

```r
typeof(42)
```

```
## [1] "double"
```

```r
typeof(42L)
```

```
## [1] "integer"
```

```r
typeof("42")
```

```
## [1] "character"
```

```r
typeof(41i)
```

```
## [1] "complex"
```

```r
typeof(TRUE)
```

```
## [1] "logical"
```

Part B: Data Frames and Conditional Statements

B1

```r
students <-data.frame(id =c(1:8),
                      name =c("Alice", "Bruno", "Chen", "Dana", "Elif", "Farid", "Grace", "Hiro"),
                      sex =c("F", "M", "M", "F", "F", "M", "F", "M"),
                      section =c("A", "B", "A", "C", "B", "C", "A", "B"),
                      midterm =c(78, 85, 92, 67, 74, 88, 95, 59),
                      final =c(82, 79, 96, 71, 80, 91, 93, 65),
                      attendance =c(95, 88, 100, 72, 90, 85, 98, 60), stringsAsFactors = FALSE)
```

B2

```r
students
```

```
##   id  name sex section midterm final attendance
## 1  1 Alice   F       A      78    82         95
## 2  2 Bruno   M       B      85    79         88
## 3  3  Chen   M       A      92    96        100
## 4  4  Dana   F       C      67    71         72
## 5  5  Elif   F       B      74    80         90
## 6  6 Farid   M       C      88    91         85
## 7  7 Grace   F       A      95    93         98
## 8  8  Hiro   M       B      59    65         60
```

```
dim(students)
```

```
## [1] 8 7
```

```
colnames(students)
```

```
## [1] "id"         "name"       "sex"        "section"    "midterm"
## [6] "final"      "attendance"
```

```
summary(students)
```

```
##        id              name               sex             section     midterm
##  Min.   :1.00   Length   :8   Length   :8   Length   :8   Min.   :59.00
##  1st Qu.:2.75   N.unique :8   N.unique :2   N.unique :3   1st Qu.:72.25
##  Median :4.50   N.blank  :0   N.blank  :0   N.blank  :0   Median :81.50
##  Mean   :4.50   Min.nchar:4   Min.nchar:1   Min.nchar:1   Mean   :79.75
##  3rd Qu.:6.25   Max.nchar:5   Max.nchar:1   Max.nchar:1   3rd Qu.:89.00
##  Max.   :8.00                                            Max.   :95.00
##      final         attendance
##  Min.   :65.00   Min.   : 60.00
##  1st Qu.:77.00   1st Qu.: 81.75
##  Median :81.00   Median : 89.00
##  Mean   :82.12   Mean   : 86.00
##  3rd Qu.:91.50   3rd Qu.: 95.75
##  Max.   :96.00   Max.   :100.00
```

B3

```
students$average <-round((students$midterm + students$final)/2, digits =1)
```

B4

```
students$passed <-students$average >=70 & students$attendance >= 75
```

B5

```
students$standing <-ifelse(students$average >=90, students$standing <-"Honors", ifelse(students$average
```

B6

```
classify <-function(score){
  if(score >=85){
  return("Distinction")
  } else if (score <=84 &score >=70) {
    return("Merit")
  } else if (score <=69 &score >=50) {
    return("Pass")
  } else if (score <50) {
    return("Fail")
  }
}
sapply(c(90, 73, 55, 40), classify)
```

```
## [1] "Distinction" "Merit"       "Pass"        "Fail"
```

Part C: rbind() and cbind()

C1

```
#new_students <-data.frame(id =c(9, 10), name =c("Ines", "Jamal"), sex =c("F", "M"), sec =c("C", "A"),
#students_all <-rbind(students, new_students)
```